Multiple Scales and Parametric Resonance

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Homework 3, Problems 3–4; original coursework excerpt

Problem 3. The children's swing

A child can amplify a swing's motion by slightly varying the swing's length periodically with a frequency that is a multiple of half the swing's natural frequency. A mathematical model for the case where the forcing frequency is equal to the natural frequency is the initial-value problem for a linear pendulum such that

$$\ddot{\phi} + (1 + \varepsilon \cos t)\phi = 0, \quad \phi(0) = 0, \ \dot{\phi}(0) = 1.$$

The slow time scale is $\tau = \varepsilon^2 t$. Pose the multi-scale expansion

$$\phi(t, \varepsilon) = \phi(t, \tau, \varepsilon) \sim \phi_0(t, \tau) + \varepsilon\phi_1(t, \tau) + \varepsilon^2\phi_2(t, \tau),$$

and find the evolution of ϕ_0 for times up to $\tau = O(1)$.

Hint: you will find that $\phi_0 = A(\tau)\cos t + B(\tau)\sin t$ and that A and B satisfy a coupled system of linear ODEs. Show that this ODE system is unstable at the fixed point A = B = 0.

Bonus: Replace the equation by

$$\ddot{\phi} + (1 + \alpha\varepsilon^2 + \varepsilon \cos t)\phi = 0, \quad \phi(0) = 0, \ \dot{\phi}(0) = 1,$$

where the constant α models a slight detuning of the pendulum from unit frequency. Adapt the multiscale analysis to this case and the range of α for which there is instability.

Solution to Problem 3

Convert ODE to PDE

Apply the chain rule:

$$\frac{d}{dt} = \partial_t + \varepsilon^2\partial_\tau, \frac{d^2}{dt^2} = \partial_t^2 + 2\varepsilon^2\partial_t\partial_\tau + O(\varepsilon^4).$$

Substitute into the equation:

$$\ddot{\phi} + (1 + \varepsilon \cos t)\phi = \Big(\partial_t^2 + 2\varepsilon^2\partial_t\partial_\tau\Big)\Big(\ \phi_0 + \varepsilon\phi_1 + \varepsilon^2\phi_2\Big) + (1 + \varepsilon \cos t)(\phi_0 + \varepsilon\phi_1 + \varepsilon^2\phi_2).$$

Order $O(1)$

$$\partial_t^2\phi_0 + \phi_0 = 0.$$

Use the "Hint". This time, we write ϕ_0 as:

$$\phi_0 = A(\tau)\cos t + B(\tau)\sin t.$$

Order $O(\varepsilon)$

$$\partial_t^2\phi_1 + \phi_1 + \cos t\phi_0 = 0.$$

Thus,

$$\partial_t^2\phi_1 + \phi_1 = -\cos t\phi_0 = -\Big(\frac{A}{2}(1+\cos 2t) + \frac{B}{2}\sin 2t\Big).$$

Solve this ODE:

$$\phi_1 = -\frac{A}{2} + \frac{A}{6}\cos 2t + \frac{B}{6}\sin 2t.$$

Order $O(\varepsilon^2)$

$$\partial_t^2\phi_2 + \phi_2 = -2\partial_t\partial_\tau\phi_0 - \cos t\phi_1.$$

Consider the right hand:

$$2\partial_t\partial_\tau\phi_0 + \cos t\phi_1 = 2\partial_\tau(-A\sin t + B\cos t) + \cos t(-\frac{A}{2} + \frac{A}{6}\cos 2t + \frac{B}{6}\sin 2t)$$

$$= -2A'\sin t + 2B\cos t + \cos t(-\frac{A}{2} + \frac{A}{6}\cos 2t + \frac{B}{6}\sin 2t).$$

Use the formula:

$$\cos x\cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)], \cos x\sin y = \frac{1}{2}[\sin(x+y) - \sin(x-y)],$$

the equation becomes:

$$(-\frac{A}{2} + 2B' + \frac{A}{12})\cos t + (-2A' + \frac{B}{12})\sin t + \frac{A}{12}\cos 3t + \frac{B}{12}\sin 3t$$

Then, we let the coefficients of resonant terms equal 0, which are $\cos t$ and $\sin t$. Then, we get a system of ODEs:

$$\boxed{-24A'(\tau) + B(\tau) = 0, 24B'(\tau) - 5A(\tau) = 0.}$$

Solve the ODE

By the initial condition,

$$\phi(0) = 0, \phi'(0) = 1 \Rightarrow \phi(0) = 0, (\partial_t + \varepsilon^2\partial_\tau)(\phi_0 + \varepsilon\phi_1 + \dots) = 0.$$

Thus,

$$\phi_0(0) = 0, \partial_t\phi_0(0) = 0 \Rightarrow A(0) = 0, B(0) = 1.$$

PS; if we write this ODE system in a matrix form:

$$\begin{pmatrix} A' \\ B' \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & \frac{1}{24} \\ \frac{5}{24} & 0 \end{pmatrix}}_{=:M} \begin{pmatrix} A \\ B \end{pmatrix}.$$

Compute the characteristic polynomial:

$$\det(M - \lambda I) = \lambda^2 - \frac{5}{576} = 0.$$

So,

$$\lambda = \pm\frac{\sqrt{5}}{24}.$$

It show that

$$(A, B) = (0, 0)$$

is an unstable point.

The final answer

Go back to the ODE system and use $A(0) = 0, B(0) = 1$. We solve ϕ_0

$$\phi_0(t.\tau) = A(\tau)\cos t + B(\tau)\sin t$$

$$= \frac{1}{\sqrt{5}}\sinh(\lambda\tau)\cos t + \cosh(\lambda\tau)\sin t,$$

where

$$\lambda = \frac{\sqrt{5}}{24}.$$

Finally,

$$\boxed{\phi(t,\varepsilon) = \frac{1}{\sqrt{5}}\sinh(\frac{\sqrt{5}}{24}\varepsilon^2 t)\cos t + \cosh(\frac{\sqrt{5}}{24}\varepsilon^2 t)\sin t}$$

Bonus

Replace the equation by

$$\ddot{\phi} + (1 + \alpha\varepsilon^2 + \varepsilon\cos t)\phi = 0, \quad \phi(0) = 0,\ \dot{\phi}(0) = 1.$$

Because of the new term $\alpha\varepsilon^2$, we have to modify our ODE system (in $(O(\varepsilon^2))$ terms):

$$2A' - (\frac{1}{12} + 2\frac{\alpha}{2})B = 0, -2B' + (\frac{5}{12} - \alpha)A = 0.$$

That is:

$$\boxed{A' = (\frac{1}{24} + \frac{\alpha}{2})B, B' = (\frac{5}{24} - \frac{\alpha}{2})A.}$$

Again, we write it in a matrix form:

$$\begin{pmatrix} A' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{24} + \frac{\alpha}{2} \\ \frac{5}{24} - \frac{\alpha}{2} & 0 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}.$$

Compute the eigenvalues:

$$\lambda^2 = (\frac{1}{24} + \frac{\alpha}{2})(\frac{5}{12} - \alpha).$$

When $\lambda^2 > 0$, there is instability.

Therefore, the range of α for which there is insability is:

$$\boxed{-\frac{1}{12} < \alpha < \frac{5}{12}.}$$

Problem 4. Kapitza's horizontal pendulum

Adapt the multi-scale computation given in class to the case of a nonlinear pendulum whose point of support is oscillating *horizontally*. The scaled governing equations are:

$$\ddot{\phi} + \sin\phi + \frac{\beta}{\varepsilon}\cos\left(\tfrac{t}{\varepsilon}\right)\cos\phi = 0.$$

Find the effective potential energy $V(\phi_0)$ for the leading-order solution and plot it for different β. Where are the stable fixed points as a function ofβ.

Solution to Problem 4

Preparation

Consider $\tau = t/\varepsilon$, use the multi-scale and expand $\phi(t,\tau)$:

$$\phi(t,\tau) \sim \phi_0(t) + \varepsilon\phi_1(t,\tau) + \varepsilon^2\phi_2(t,\tau).$$

Find partial derivatives:

$$\frac{d}{dt} = \partial_t + \frac{1}{\varepsilon}\partial_\tau, \qquad \frac{d^2}{dt^2} = \partial_t^2 + \frac{2}{\varepsilon}\partial_t\partial_\tau + \frac{1}{\varepsilon^2}\partial_\tau^2.$$

Substitute into the equation:

$$\ddot{\phi} + \sin\phi + \frac{\beta}{\varepsilon}\cos\left(\tfrac{t}{\varepsilon}\right)\cos\phi$$

$$= (\partial_t^2 + \frac{2}{\varepsilon}\partial_t\partial_\tau + \frac{1}{\varepsilon^2}\partial_\tau^2)(\phi_0 + \varepsilon\phi_1 + \varepsilon^2\phi_2) + \sin\phi + \frac{\beta}{\varepsilon}\cos(\tau)\cos\phi$$

Then, expand $\sin\phi$ and $\cos\phi$:

$$\sin(\phi) \sim \sin(\phi_0 + \varepsilon\phi_1 + \varepsilon^2\phi_2) = \sin\phi_0 + \varepsilon\phi_1\cos\phi_0 + O(\varepsilon^2),$$

$$\frac{\beta}{\varepsilon}\cos(\tau)\cos\phi = \frac{\beta}{\varepsilon}\cos(\tau)\Big(\cos\phi_0 - \varepsilon\phi_1\sin\phi_0 + \dots\Big).$$

Notice that ϕ_0 does not depend on τ.(By the assumption of the expansion) So,

$$\partial_\tau\phi_0 = 0.$$

Order $O(\frac{1}{\varepsilon^2})$

Since

$$\partial_\tau\phi_0 = 0,$$

the Order $O(\frac{1}{\varepsilon^2})$ term equal 0.

Order $O(\frac{1}{\varepsilon})$

$$\partial_\tau^2\phi_1 + 2\partial_t\partial_\tau\phi_0 = -\beta\cos\tau\cos\phi_0,$$

Because $\partial_\tau\phi_0 = 0$,

$$2\partial_t\partial_\tau\phi_0 = 0.$$

Thus,

$$\partial_\tau^2\phi_1 = -\beta\cos\tau\cos\phi_0.$$

To solve this PDE, we have to find the general solution and periodic special solution.

The general solution The solution of $\partial_\tau^2\phi_1 = 0$ is $a(t)\tau + b(t)$. But we know that ϕ_1 has nothing to do with τ. Thus, $b(t) = 0$. Then, we pick $a(t) = 0$.

The special solution Assume the special solution is $A(t)\cos(\tau)$. Substitute the special solution and get $A(t) = \beta\cos\phi_o(t)$.

Therefore,

$$\boxed{\phi_1(t,\tau) = \beta\cos\phi_0(t)\cos\tau.}$$

Order $O(1)$

Collect all the terms:

$$\partial_t^2\phi_0 + 2\partial_t\partial_\tau\phi_1 + \partial_\tau^2\phi_2 + \sin\phi_0 - \beta\cos\tau\phi_1\sin\phi_0 = 0.$$

We can't cancel the resonant terms here, so we average the $O(1)$ equation in τ. Notice that $\overline{\partial_\tau} = 0$:

$$\partial_t^2\phi_0 + \sin\phi_0 - \frac{\beta^2}{2}\cos\phi_0\sin\phi_0 = 0.$$

This can be written in the form $\partial_t^2\phi_0 + \dfrac{dV}{d\phi_0} = 0$

$$\boxed{V(\phi_0) = -\cos\phi_0 - \frac{\beta^2}{4}\sin^2\phi_0\,.}$$

Let $V'(\phi_0) = 0$. Thus,

- $\phi = 0, \pi$;
- when $\beta^2 \geq 2$, two solutions with

$$\cos\phi = \frac{2}{\beta^2}, \qquad \phi = \pm\arccos\Big(\frac{2}{\beta^2}\Big).$$

Stable fixed points are determined by

$$V''(\phi) = \cos\phi - \tfrac{\beta^2}{2}(2\cos^2\phi - 1).$$

If $V''(\phi) > 0$, it is stable.

- At $\phi = 0$:

$$V''(0) = 1 - \frac{\beta^2}{2}.$$

  Hence $\phi = 0$ is stable if $1 - \frac{\beta^2}{2} > 0$.

- At $\phi = \pi$:

$$V''(\pi) = -1 - \frac{\beta^2}{2} < 0,$$

  so $\phi = \pi$ is always unstable for all values of β.

- For the additional equilibria

$$\cos\phi = \frac{2}{\beta^2}, \quad \text{(which exist only if } \beta^2 \geq 2\text{)}.$$

  So,

$$V'' = \frac{\beta^4 - 4}{2\beta^2}.$$

  Thus, when $\beta^2 > 2$, these equilibria are stable since $V'' > 0$. At $\beta^2 = 2$ they degenerate with $V'' = 0$.(corresponding to a bifurcation point)

Summary:

- For $0 \leq \beta < \sqrt{2}$: the only stable equilibrium is $\boxed{\phi = 0}$.
- For $\beta = \sqrt{2}$: $\phi = 0$ is "marginally" stable.
- For $\beta > \sqrt{2}$: $\phi = 0$ is unstable, and two symmetric stable equilibria appear at

$$\boxed{\phi = \pm \arccos\left(\frac{2}{\beta^2}\right)}.$$

  $\phi = \pi$ is always unstable.

Plot

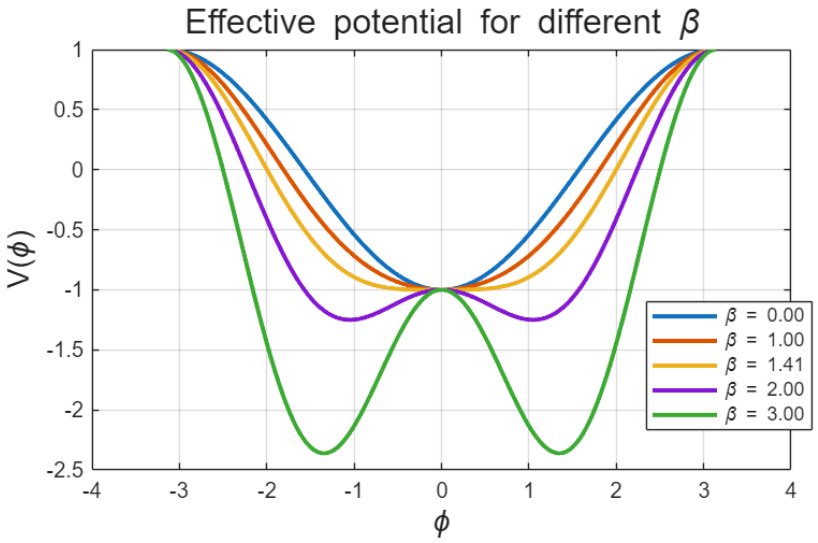


Figure 1: Effective potential $V(\phi)$ for different values of β.